Data Appendix: Metro-Level AI Exposure Index Construction

1 Data Sources

1.1 IPUMS American Community Survey

Individual-level microdata were obtained from IPUMS USA (`https://usa.ipums.org`). The extract (`usa_00002`) draws on the **2019 single-year American Community Survey (ACS)** (sample code `201903`). The extract contains 7,818,941 person records with the following variables: `YEAR`, `MULTYEAR`, `SAMPLE`, `SERIAL`, `CBSERIAL`, `HHWT`, `CLUSTER`, `MET2013`, `STRATA`, `GQ`, `PERNUM`, `PERWT`, `EMPSTAT`, `EMPSTATD`, `OCCSOC`, and `INCWAGE`.

After dropping records with missing occupation codes (`OCCSOC`), 6,840,674 person records remain, containing 471 unique occupation codes. Occupation codes in IPUMS ACS are reported using the **2018 Standard Occupational Classification (SOC)** system.

Metropolitan areas are identified by the `MET2013` variable, which uses 2013 Office of Management and Budget (OMB) metropolitan area delineations.

1.2 Felten et al. AI Exposure Scores

Occupation-level AI exposure scores are from felten2021.[1] The published dataset contains 1,430 rows covering **715 unique occupations** coded in the **2010 SOC** system. Of these, 714 are detailed-level codes; one (19-1020) is a broad-level code.

1.3 BLS SOC Crosswalks

Two reference files from the Bureau of Labor Statistics were used:

1. **2010 SOC to 2018 SOC crosswalk** (`soc_2010_to_2018_crosswalk.xlsx`). Contains 901 mapping rows linking 841 unique 2010 SOC codes to 868 unique 2018 SOC codes.
2. **2018 SOC structure file** (`2018soc.xlsx`). Provides the full hierarchy of the 2018 SOC: 23 Major groups, 98 Minor groups, 459 Broad groups, and 867 Detailed groups.

[1] Felten, E. W., Raj, M., & Seamans, R. (2021). Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses. *Strategic Management Journal*, 42(12), 2195–2217.

2 SOC Crosswalk: 2018 to 2010

Because the ACS microdata use 2018 SOC codes while the Felten et al. scores use 2010 SOC codes, a reverse crosswalk from 2018 to 2010 was constructed from the BLS official crosswalk. Table 1 summarises the mapping structure.

Table 1: BLS 2010–2018 SOC Crosswalk Summary

Statistic	**Count**
Total mapping rows	901
Unique 2010 SOC codes	841
Unique 2018 SOC codes	868
1:1 mappings (one 2018 → one 2010)	837
Many-to-one (one 2018 → multiple 2010)	31
One-to-many (one 2010 → multiple 2018)	42

The crosswalk is not one-to-one in all cases because the 2018 SOC revision split some 2010 occupations and merged others. Of the 868 unique 2018 codes in the crosswalk, 837 (96.4%) map to exactly one 2010 code, while 31 (3.6%) map to two or more 2010 codes.

3 Matching ACS Occupation Codes to Felten Scores

3.1 Procedure

IPUMS reports some occupation codes at aggregated levels (Broad, Minor, or Major groups) rather than the Detailed level used by the crosswalk. The matching proceeds in two stages:

1. **Direct lookup.** If the 2018 code appears in the crosswalk as a detailed code, retrieve the corresponding 2010 code(s).
2. **Hierarchical expansion.** If the code is a Broad, Minor, or Major group code, expand it to its constituent Detailed codes using the 2018 SOC structure file, then look up each Detailed code in the crosswalk.

After obtaining the set of 2010 SOC codes for each 2018 code, the corresponding Felten AI exposure scores are retrieved.

3.2 Score Assignment Rules

- **1:1 match** (single Felten score found): assign that score directly.
- **Ambiguous match** (multiple Felten scores found): assign the *equal-weighted arithmetic mean* of all matched scores.
- **No match**: the observation is retained with a missing (`NaN`) exposure score.

3.3 Match Results

Of the 471 unique 2018 SOC codes observed in the ACS extract:

Table 2: Occupation Code Match Outcomes

Category	**Codes**	**Share (%)**
1:1 match (single Felten score)	349	74.1
Ambiguous (averaged over multiple scores)	86	18.3
No Felten match (2010 code not in Felten)	34	7.2
Not in crosswalk or SOC structure	2	0.4
Total	**471**	**100.0**

At the person-record level, 6,240,864 of 6,840,674 records (91.2%) receive a valid AI exposure score. Weighted by `PERWT`, 132.1 million of 144.4 million workers (91.5%) are covered.

4 Ambiguous Matches: Codes Averaged Over Multiple Felten Scores

Of the 86 ambiguous codes, ambiguity arises from two distinct sources:

1. **Direct crosswalk ambiguity** (10 codes). A detailed 2018 SOC code maps to multiple 2010 SOC codes in the BLS crosswalk, each of which has a Felten score (Table 3).
2. **Hierarchical expansion ambiguity** (76 codes). IPUMS reports the occupation at the Broad, Minor, or Major group level. Expanding to constituent Detailed codes and crosswalking to 2010 yields multiple distinct Felten scores (Table 4).

In both cases, the assigned AI exposure score is the equal-weighted arithmetic mean of all matched Felten scores.

4.1 Panel A: Direct Crosswalk Ambiguity (10 codes)

Table 3: Direct Crosswalk Ambiguous Matches (Detailed 2018 SOC → Multiple 2010 Felten Codes)

2018 SOC	**Title**	**Mapped 2010 codes**	N
13-1082	Project Management Specialists	11-9199, 13-1199, 15-1199	3
15-1252	Software Developers	15-1132, 15-1133	2
15-1253	Software QA Analysts and Testers	15-1132, 15-1133, 15-1199	3
15-1255	Web and Digital Interface Designers	15-1134, 15-1199	2
15-1299	Computer Occupations, All Other	15-1199, 43-9011	2
25-4022	Librarians and Media Collections Spec.	25-4021, 25-9011	2
27-3023	News Analysts, Reporters, Journalists	27-3021, 27-3022	2

2018 SOC	Title	Mapped 2010 codes	*N*
45-3031	Fishing and Hunting Workers	45-3011, 45-3021	2
47-5032	Explosives Workers, Ordnance Handling	47-5021, 47-5031	2
53-3053	Shuttle Drivers and Chauffeurs	53-3022, 53-3041	2

4.2 Panel B: Hierarchical Expansion Ambiguity (76 codes)

These 76 codes are reported by IPUMS at an aggregated level (Broad, Minor, or Major). After expanding to Detailed 2018 codes and crosswalking to 2010, each maps to multiple distinct Felten scores.

Table 4: Expanded Ambiguous Matches (Aggregated 2018 SOC → Multiple 2010 Felten Codes)

2018 SOC	Title	Level	*N*
11-9030	Education and Childcare Administrators	Broad	3
11-9070	Entertainment and Recreation Managers	Broad	2
13-1030	Claims Adjusters, Appraisers, Examiners, and Investigators	Broad	2
13-1070	Human Resources Workers	Broad	3
13-2070	Credit Counselors and Loan Officers	Broad	2
15-1230	Computer Support Specialists	Broad	2
17-1020	Surveyors, Cartographers, and Photogrammetrists	Broad	2
17-2070	Electrical and Electronics Engineers	Broad	2
19-1010	Agricultural and Food Scientists	Broad	3
19-1020	Biological Scientists	Broad	4
19-1030	Conservation Scientists and Foresters	Broad	2
19-2010	Astronomers and Physicists	Broad	2
19-2030	Chemists and Materials Scientists	Broad	2
19-5010	Occup. Health and Safety Specialists and Technicians	Broad	2
25-1000	Postsecondary Teachers	Minor	34
25-2010	Preschool and Kindergarten Teachers	Broad	2
25-2020	Elementary and Middle School Teachers	Broad	2
25-2050	Special Education Teachers	Broad	4
25-4010	Archivists, Curators, and Museum Technicians	Broad	3
25-9040	Teaching Assistants	Broad	2
27-1010	Artists and Related Workers	Broad	4
27-2030	Dancers and Choreographers	Broad	2
27-4030	TV, Video, and Film Camera Operators and Editors	Broad	2
29-1020	Dentists	Broad	4
29-1210	Physicians	Broad	7
29-1240	Surgeons	Broad	2
29-2010	Clinical Laboratory Technologists and Technicians	Broad	2
29-2090	Misc. Health Technologists and Technicians	Broad	4
29-9000	Other Healthcare Practitioners and Technical Occ.	Minor	4

2018 SOC	Title	Level	N
31-2010	Occupational Therapy Assistants and Aides	Broad	2
31-2020	Physical Therapist Assistants and Aides	Broad	2
33-2020	Fire Inspectors	Broad	2
33-3050	Police Officers	Broad	2
33-9030	Security Guards and Gambling Surveillance Officers	Broad	2
35-2010	Cooks	Broad	5
39-1000	Supervisors of Personal Care and Service Workers	Minor	3
39-3010	Gambling Services Workers	Broad	2
39-6010	Baggage Porters, Bellhops, and Concierges	Broad	2
39-7010	Tour and Travel Guides	Broad	2
41-2010	Cashiers	Broad	2
41-9010	Models, Demonstrators, and Product Promoters	Broad	2
41-9020	Real Estate Brokers and Sales Agents	Broad	2
45-4020	Logging Workers	Broad	3
47-2040	Carpet, Floor, and Tile Installers and Finishers	Broad	4
47-2050	Cement Masons, Concrete Finishers, and Terrazzo Workers	Broad	2
47-2070	Construction Equipment Operators	Broad	2
47-2080	Drywall Installers, Ceiling Tile Installers, and Tapers	Broad	2
47-2130	Insulation Workers	Broad	2
47-2140	Painters and Paperhangers	Broad	2
47-3010	Helpers, Construction Trades	Broad	5
47-5020	Surface Mining Machine Operators and Earth Drillers	Broad	2
47-5040	Underground Mining Machine Operators	Broad	5
49-3040	Heavy Vehicle and Mobile Equipment Service Technicians	Broad	3
49-3050	Small Engine Mechanics	Broad	3
49-3090	Misc. Vehicle and Mobile Equipment Mechanics	Broad	3
49-9060	Precision Instrument and Equipment Repairers	Broad	4
51-2020	Electrical, Electronics, and Electromechanical Assemblers	Broad	3
51-3020	Butchers and Other Meat, Poultry, and Fish Processing	Broad	3
51-4050	Metal Furnace Operators, Tenders, Pourers, and Casters	Broad	2
51-6040	Shoe and Leather Workers	Broad	2
51-6050	Tailors, Dressmakers, and Sewers	Broad	2
51-6060	Textile Machine Setters, Operators, and Tenders	Broad	2
51-8010	Power Plant Operators, Distributors, and Dispatchers	Broad	3
51-8090	Misc. Plant and System Operators	Broad	3
51-9020	Crushing, Grinding, Polishing, Mixing, and Blending Workers	Broad	2
51-9030	Cutting Workers	Broad	2
51-9080	Dental and Ophthalmic Lab Technicians and Med. Appliance Tech.	Broad	3
51-9120	Painting Workers	Broad	2
51-9160	CNC Tool Operators and Programmers	Broad	2
53-1000	Supervisors of Transportation and Material Moving	Minor	2
53-2010	Aircraft Pilots and Flight Engineers	Broad	2
53-2020	Air Traffic Controllers and Airfield Operations Spec.	Broad	2

2018 SOC	Title	Level	N
53-3030	Driver/Sales Workers and Truck Drivers	Broad	3
53-4010	Locomotive Engineers and Operators	Broad	2
53-5020	Ship and Boat Captains and Operators	Broad	2
53-7070	Pumping Station Operators	Broad	3

5 Unmatched Occupation Codes

A total of 36 ACS occupation codes receive no Felten AI exposure score. These fall into two groups depending on the reason for non-matching.

5.1 No Felten Match (34 codes)

These 34 codes can be mapped to 2010 SOC codes through the crosswalk (directly or via hierarchical expansion), but the resulting 2010 codes are absent from the Felten dataset. Common reasons include "All Other" catch-all categories, first-line supervisor codes, military occupations, and occupations created or reorganised in the 2018 SOC revision that have no clear 2010 antecedent in the Felten data.

Table 5: 2018 SOC Codes with No Felten Match—In Crosswalk but 2010 Code Not in Felten ($n = 34$)

SOC Code	Title
13-1011	Agents and Business Managers of Artists, Performers, and Athletes
21-1019	Counselors, All Other
21-1029	Social Workers, All Other
21-2099	Religious Workers, All Other
23-2099	Legal Support Workers, All Other
27-2091	Disc Jockeys, Except Radio
27-2099	Entertainers and Performers, Sports and Related Workers, All Other
33-1090	Misc. First-Line Supervisors, Protective Service Workers
35-3023	Fast Food and Counter Workers
35-9099	Food Preparation and Serving Related Workers, All Other
37-1012	First-Line Supervisors of Landscaping, Lawn Service, and Groundskeeping
39-9099	Personal Care and Service Workers, All Other
41-4010	Sales Representatives, Wholesale and Manufacturing
41-9091	Door-to-Door Sales Workers, News and Street Vendors, and Related
41-9099	Sales and Related Workers, All Other

SOC Code	Title
43-1011	First-Line Supervisors of Office and Admin. Support Workers
43-5053	Postal Service Mail Sorters, Processors, and Processing Machine Operators
43-6014	Secretaries and Admin. Assistants, Except Legal, Medical, and Executive
47-1011	First-Line Supervisors of Construction Trades and Extraction Workers
49-2097	Audiovisual Equipment Installers and Repairers
49-9021	Heating, AC, and Refrigeration Mechanics and Installers
51-3091	Food and Tobacco Roasting, Baking, and Drying Machine Operators
51-3099	Food Processing Workers, All Other
51-4020	Forming Machine Setters, Operators, and Tenders, Metal and Plastic
51-4031	Cutting, Punching, and Press Machine Setters, Metal and Plastic
51-4033	Grinding, Lapping, Polishing, and Buffing Machine Tool Setters, Metal and Plastic
51-7042	Woodworking Machine Setters, Operators, and Tenders, Except Sawing
51-9041	Extruding, Forming, Pressing, and Compacting Machine Setters
51-9151	Photographic Process Workers and Processing Machine Operators
53-3011	Ambulance Drivers and Attendants, Except EMTs
53-3099	Motor Vehicle Operators, All Other
55-1010	Military Officer Special and Tactical Operations Leaders
55-2010	First-Line Enlisted Military Supervisors
55-3010	Military Enlisted Tactical Operations and Air/Weapons Specialists

5.2 Not in Crosswalk or SOC Structure (2 codes)

These 2 codes are IPUMS-specific classifications with no standard SOC equivalent and therefore cannot be crosswalked to 2010.

Table 6: 2018 SOC Codes Not in Crosswalk or SOC Structure ($n = 2$)

SOC Code	Title
55-9830	Military, Rank Not Specified (IPUMS)
99-9920	Unemployed or Last Worked $>$ 5 Years Ago (IPUMS)

6 Metro-Level Aggregation

6.1 Aggregation Method

For each metropolitan area m identified by `MET2013`, the AI exposure index is computed as a person-weight–weighted average:

$$\text{AI Exposure}_m = \frac{\sum_{i \in m} w_i \cdot s_i}{\sum_{i \in m} w_i}, \tag{1}$$

where w_i is the ACS person weight (`PERWT`) and s_i is the Felten AI exposure score for individual i. Only individuals with a valid (non-missing) AI exposure score are included in the numerator and denominator.

6.2 Coverage

Table 7: Metro-Level Coverage Summary

Statistic	**Value**
Number of metropolitan areas	261
Overall weighted coverage (`PERWT`)	91.5%
Minimum metro coverage	80.4%
Maximum metro coverage	93.7%
Mean metro coverage	91.3%
Metros with $\geq$ 90% coverage	240 (91.9%)
Metros with $\geq$ 95% coverage	0

Missing scores are concentrated in supervisor, military, and miscellaneous "All Other" occupation categories that are distributed fairly uniformly across metros, so variation in coverage across metros is small (range: 80.4–93.7%).

6.3 AI Exposure Distribution

Table 8: Distribution of Metro-Level AI Exposure

Statistic	**Value**
Minimum	−0.465
25th percentile	—
Median	−0.205
Mean	−0.199
Maximum	0.232

7 Summary of Methodological Choices

1. **SOC vintage mismatch.** The ACS microdata use 2018 SOC codes; the Felten et al. scores use 2010 SOC codes. We bridge this gap using the BLS official 2010-to-2018 SOC crosswalk (901 mapping rows).

2. **Aggregated IPUMS codes.** IPUMS sometimes reports occupation at the Broad, Minor, or Major group level rather than the Detailed level. These are expanded to their constituent Detailed codes via the 2018 SOC structure file before crosswalking to 2010 codes.

3. **Ambiguous crosswalk matches.** When a 2018 code maps to multiple 2010 codes with Felten scores (86 codes; see Tables 3 and 4), we assign the equal-weighted average of all matched scores. Of these, 10 arise from direct many-to-many mappings in the BLS crosswalk and 76 from hierarchical expansion of IPUMS Broad, Minor, or Major group codes to their constituent Detailed codes. An employment-weighted average across the mapped 2010 occupations would be a natural alternative but requires auxiliary employment data at the occupation level.

4. **Unmatched occupations.** 36 of 471 ACS occupation codes (7.6%) receive no AI exposure score: 34 codes have a crosswalk mapping but the resulting 2010 code is absent from the Felten dataset (Table 5), and 2 are IPUMS-specific codes with no SOC equivalent (Table 6). These account for 8.5% of weighted employment. The unmatched codes are primarily military occupations, first-line supervisor aggregates, and "All Other" residual categories. These observations are excluded from the metro-level weighted average but are retained in coverage statistics.

5. **Weighting.** Metro-level exposure is the `PERWT`-weighted mean across individuals, reflecting each person's representation in the population.

6. **Metropolitan area definition.** We use the `MET2013` variable from IPUMS, which corresponds to 2013 OMB metropolitan area delineations. This yields 261 identifiable metropolitan areas.